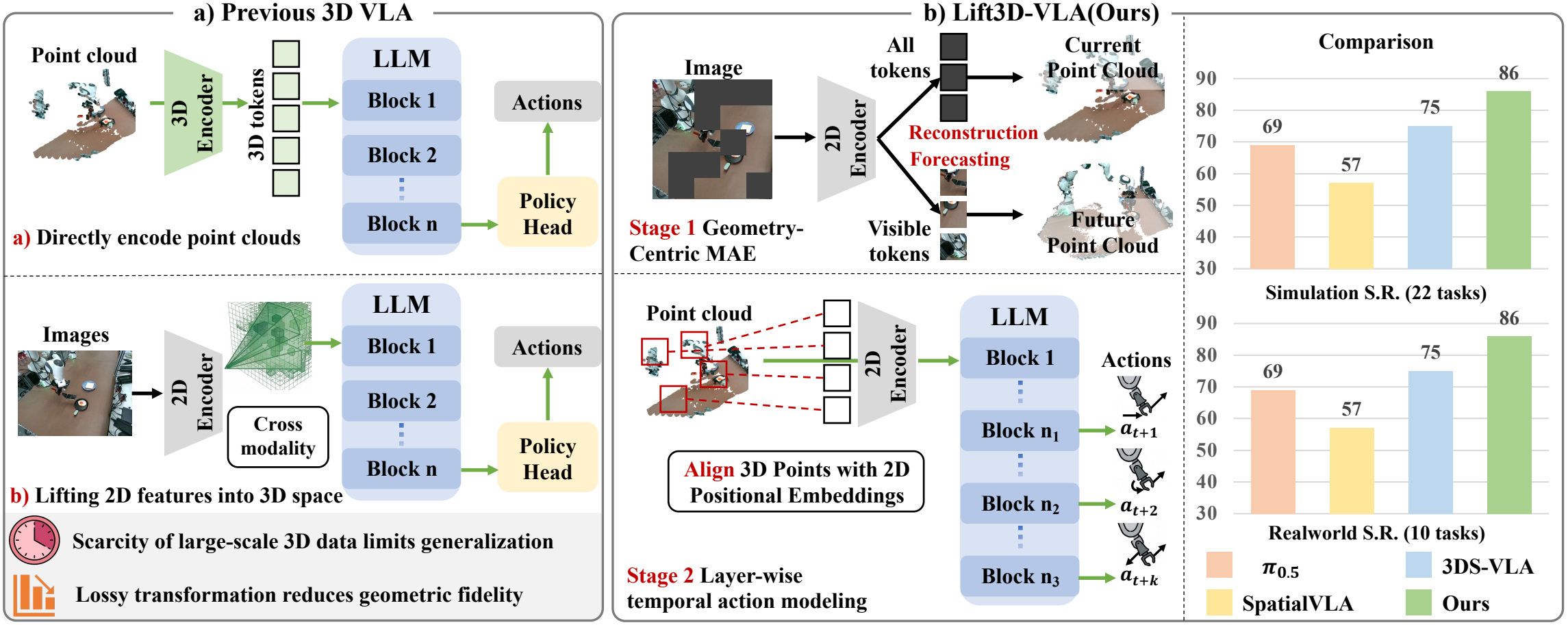
a) Previous 3D VLA
Point cloud
3D Encoder
3D tokens
LLM
Block 1
Block 2
Block n
Actions
Policy Head
a) Directly encode point clouds
Images
2D Encoder
Cross modality
LLM
Block 1
Block 2
Block n
Actions
Policy Head
b) Lifting 2D features into 3D space
Scarcity of large-scale 3D data limits generalization
Lossy transformation reduces geometric fidelity
b) Lift3D-VLA(Ours)
Image
2D Encoder
All tokens
Reconstruction
Forecasting
Visible tokens
Current Point Cloud
Future Point Cloud
Stage 1 Geometry-Centric MAE
Point cloud
2D Encoder
Align 3D Points with 2D Positional Embeddings
LLM
Block 1
Block n1
Block n2
Block n3
Actions
a_{t+1}
a_{t+2}
a_{t+k}
Stage 2 Layer-wise temporal action modeling
Comparison
90
80
70
60
50
40
30
69
57
75
86
Simulation S.R. (22 tasks)
Realworld S.R. (10 tasks)
$\pi_{0.5}$
3DS-VLA
SpatialVLA
Ours